

Composition

AVORAX 200mg: Each tablet contains Acyclovir 200mg.
AVORAX 400mg: Each tablet contains Acyclovir 400mg.

Pharmacodynamics

Acyclovir is a synthetic purine nucleoside analogue with in vitro and in vivo inhibitory activity against human herpes viruses, including herpes simplex virus (HSV) types I and II and varicella zoster virus (VZV). The inhibitory activity of acyclovir for HSV I, HSV II and VZV is highly selective. The enzyme thymidine kinase (TK) of normal, uninfected cells does not use acyclovir effectively as a substrate, hence toxicity of mammalian host cells is low; however, TK encoded by HSV and VZV converts acyclovir to acyclovir monophosphate, a nucleoside analogue which is further converted to the diphosphate and finally to the triphosphate by cellular enzymes. Acyclovir triphosphate interferes with the viral DNA polymerase and inhibits viral DNA replication with resultant chain termination following its incorporation into the viral DNA. Prolonged or repeated courses of acyclovir in severely immune-compromised individuals may result in the selection of virus strains with reduced sensitivity, which may not respond to continued acyclovir treatment. Most of the clinical isolates with reduced sensitivity have been relatively deficient in viral TK, however, strains with altered viral TK or viral DNA polymerase have also been reported. In vitro exposure of HSV isolates to acyclovir can also lead to the emergence of less sensitive strains. The relationship between the in vitro determined sensitivity of HSV isolates and clinical response to acyclovir therapy is not clear.

Pharmacokinetics

Acyclovir is only partially absorbed from the gut. Mean steady state peak plasma concentrations (C_{max}) following doses of 200 mg administered four-hourly were 3.1 microMol (0.7 micrograms/ml) and equivalent trough plasma levels (C_{min}) were 1.8 microMol (0.4 micrograms/ml). Corresponding C_{max} levels following doses of 400 mg and 800 mg administered four-hourly were 5.3 microMol (1.2 micrograms/ml) and 8 microMol (1.8 micrograms/ml) respectively and equivalent C_{min} levels were 2.7 microMol (0.6 micrograms/ml) and 4 microMol (0.9 micrograms/ml). Most of the drug is excreted unchanged by the kidney. Renal clearance of acyclovir is substantially greater than creatinine clearance, indicating that tubular secretion, in addition to glomerular filtration contributes to the renal elimination of the drug. 9-carboxymethoxymethylguanine is the only significant metabolite of acyclovir, and accounts for approximately 10 - 15% of the administered dose recovered from the urine. When acyclovir is given one hour after 1 gram of probenecid the terminal half-life and the area under the plasma concentration time curve is extended by 18% and 40% respectively. In the elderly, total body clearance falls with increasing age associated with decreases in creatinine clearance although there is little change in the terminal plasma half-life. In patients with chronic renal failure the mean terminal half-life was found to be 19.5 hours. The mean acyclovir half-life during haemodialysis was 5.7 hours. Plasma acyclovir levels dropped approximately 60% during dialysis. Cerebrospinal fluid levels are approximately 50% of corresponding plasma levels. Plasma protein binding is relatively low (9 to 33%) and drug interactions involving binding site displacement are not anticipated.

Indications

Avorax is indicated in the treatment of viral infections due to herpes simplex virus (type 1 & 2) and varicella zoster (chickenpox) and herpes zoster infections. Herpes simplex infections zoster including herpes keratitis, herpes labialis and genital herpes. Both initial and recurrent herpes simplex infections can be successfully treated. Prolonged treatment can reduce the incidence of recurrence which is important in immuno-compromised patients.

Dosage and Administration

For oral administration only.

Dosage in Adults:

For treatment of herpes simplex infections:

200mg should be taken 5 times daily at approximately four hourly intervals for 5 to 10 days. In severely immunocompromised patients (e.g. after marrow transplant) or in patients with impaired absorption from the gut, the dose can be doubled to 400mg.

Dosing should begin as early as possible after the start of an infection; for recurrent episode this should preferably be during the prodromal period or when the lesions first appear.

For suppression of recurrent herpes simplex in the immunocompetent patients:

Starting dose of 200mg should be taken four times daily at approximately six-hourly intervals. Many patients may be conveniently managed on a regimen of 400mg, twice daily at approximately 12-hourly intervals. Dosage titration down to 200mg 2 or 3 times daily may prove effective. Some patients may experience break-through infections on total daily doses of 800mg. Therapy should be interrupted periodically at intervals of 6 to 12 months, in order to reassess suitability for continued suppression.

For prophylaxis herpes simplex infections in the immunocompromised patients:

200mg should be taken FOUR times daily at approximately six hourly intervals.

In severely immunocompromised patients (e.g. after marrow transplant) or in patients with impaired absorption from the gut, the dose can be doubled to 400mg or alternatively, intravenous dosing could be considered. The duration of prophylactic administration is determined by the duration of the period at risk.

For treatment of Herpes Zoster (shingles):

800mg 5 times daily for 7 - 10 days.

Dosage in children:

For treatment of herpes simplex infections and for prophylaxis of herpes simplex infections in the immunocompromised patients:

Children over the age of 2 years should be given adult dosages and children below the age of 2 years should be given half the adult dose. No specific data are available on the suppression of herpes simplex infections in immunocompetent children.

For treatment of varicella infections (chickenpox):

Children 2 - 12 years: 20mg/kg four times daily (up to a maximum of 800mg per dose) for 5 days.

Children 13 -18 years: 800mg four times daily for 5 days.

Dosage adjustment in Renal Impairment:

For patient with a creatinine level of less than 10 ml per minute, oral dose need to be reduced to 200mg every 12 hours for herpes simplex and 800mg every 12 hours for varicella-zoster infections. Dose reduction might also be required in the elderly.

Dosage in the elderly:

In the elderly, total body clearance declines in parallel with creatinine clearance. Adequate hydration of elder patients taking high oral doses of Acyclovir should be maintained. Special attention should be given to dosage reduction in the elderly patients with impaired renal function.

Contraindications

It is contraindicated in patients with known hypersensitivity to Acyclovir.

Precautions

Acyclovir should be given with caution to elderly patients and patients with renal failure. The doses must be reduced according to the creatinine clearance.

Also use with caution in dehydrated patients. The risk of renal impairment is higher in dehydrated patients.



Use in pregnancy & lactation

Limited data are available on the use of acyclovir during pregnancy. Caution should therefore be exercised by balancing the potential benefits of treatment against any possible hazard to the foetus. There is some evidence that the drug does pass into breast milk. Caution is therefore advised if acyclovir is to be administered to a nursing mother.

Drug Interactions

Extreme lethargy is usually associated when used with zidovudine. Probenecid reduces acyclovir excretion, hence increase acyclovir plasma concentrations.

Incompatibilities: None known

Overdosage and Treatment

Avorax is relatively non-toxic. However, in the event of severe overdosage, supportive and symptomatic treatment can be given.

Renal impairment responds rapidly to hydration and/or reduction in the dosage. If necessary, haemodialysis can be performed.

Side Effects

Side effects are headache, neurological reactions and fatigue. Skin rashes and gastrointestinal disturbance may occur but is well tolerated on oral administration.

Other side-effects are increase in blood urea, creatinine level, bilirubin and liver-related enzymes and haematological indices. There are also reports of confusions, tremors, seizures especially in the immunosuppressed patients.

Presentation

Avorax 200mg: Blue colour, round tablet with an inverted pyramid logo embossed on one side, plain on the other side, 12.5 mm in diameter in blister packs of 25's.

Avorax 400mg: Pink colour, round tablet with an inverted pyramid logo embossed on one side, plain on the other side, 12.5 mm in diameter in blister packs of 25's.

Shelf-Life: The expiry date is indicated on the packaging.

Storage

Protect from light. Store in a cool, dry place below 30°C.

KEEP OUT REACH OF CHILDREN

JAUHI DARI KANAK-KANAK

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Manufacturer and Product Registration Holder

Xepa-Soul Pattinson (Malaysia) Sdn Bhd

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